

feedback — fusion

feedback — GNN

## NF-UNSW-NB15

|                     |         |         |
|---------------------|---------|---------|
| Normal (311)        | +0.0390 | +0.0414 |
| Analysis (25)       | +0.0309 | +0.1869 |
| Backdoors (40)      | +0.1442 | +0.2090 |
| DoS (40)            | +0.0978 | +0.1594 |
| Exploits (40)       | −0.0003 | +0.0271 |
| Fuzzers (40)        | +0.0419 | +0.1542 |
| Generic (40)        | +0.1490 | +0.0308 |
| Reconnaissance (40) | −0.0185 | −0.0379 |
| Shellcode (40)      | +0.0531 | +0.1290 |
| Worms (40)          | +0.0107 | +0.0040 |

## NF-ToN-IoT

|                 |         |         |
|-----------------|---------|---------|
| Benign (1746)   | +0.0555 | +0.0237 |
| backdoor (16)   | +0.2280 | +0.0903 |
| ddos (35)       | +0.0502 | +0.1099 |
| injection (116) | +0.1427 | +0.0842 |
| mitm (157)      | +0.1484 | −0.0273 |
| password (25)   | −0.0862 | +0.1733 |
| scanning (13)   | +0.0382 | +0.2242 |
| xss (12)        | +0.1768 | −0.1116 |

difference in per-class F1

Three-seed means. Class sizes in brackets. A black outline marks a loss. No per-class difference carries an interval, and on classes of 12 to 40 edges a single edge moves F1 substantially.